

Executive Summary

Workforce Attrition Analytics as an Early-Warning Signal for Operational Risk

Context

In supply chain and logistics environments, operational performance depends heavily on the availability and stability of skilled labor. Workforce attrition is often treated as a human-resources issue, yet in practice it represents a **material operational risk**, leading to capacity shortages, service disruption, increased overtime costs, and reactive decision-making.

This project reframes workforce attrition analytics as a **leading indicator of operational risk**, rather than a backward-looking HR metric.

Business Problem

Operations leaders typically become aware of workforce-related capacity issues **after** performance has already deteriorated (missed deliveries, delays, emergency subcontracting). Traditional HR reporting does not provide early visibility into which workforce segments pose elevated operational risk.

Key questions addressed:

- Which workforce segments exhibit elevated attrition risk?
- Can attrition drivers be identified **before** they translate into operational disruption?
- How can people data support proactive supply-chain decision-making?

Objective

To develop an analytics-driven framework that:

- Identifies workforce segments with elevated attrition risk
- Interprets people-related signals through an **operational lens**
- Supports proactive workforce and capacity planning decisions

Approach

- Exploratory data analysis to identify attrition patterns across roles, tenure, and compensation
- Feature engineering to capture employment stability and engagement proxies
- Classification modeling to highlight workforce segments associated with elevated risk
- Interpretation focused on **decision support**, not model performance

The emphasis is on **actionability**, not predictive accuracy alone.

Key Insights

- Attrition risk is unevenly distributed across specific roles and tenure bands
- Workforce volatility can be detected **before** operational KPIs deteriorate
- People-related indicators can function as early warning signals for capacity risk
- Analytics enables targeted intervention rather than reactive hiring

Operational Impact

When integrated into supply-chain governance, this approach can help:

- Anticipate capacity shortfalls before service levels are impacted
- Reduce unplanned disruptions linked to workforce instability
- Improve workforce planning for critical logistics roles
- Support structured escalation of operational risk to leadership

Decision Ownership

This analysis is designed to support:

- Operations Directors responsible for capacity and service continuity
- Supply Chain Managers managing execution risk
- HR & Operations leadership coordinating workforce planning

Limitations & Next Steps

- Analysis based on a synthetic dataset
- Next steps include integration with operational KPIs such as OTIF, productivity, and overtime
- Extension into multi-site dashboards for executive monitoring